3D Vectors & the Cross Product

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, 2)$ and $B(4, 6)$. Find the vector $\vec{AB}$.

Final Answer: $(3; 4)$. To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. \\ Given $A(x_1, y_1) = (1, 2)$ and $B(x_2, y_2) = (4, 6)$. \\ The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1)$. \\ Substituting the given coordinates: \\ $\vec{AB} = (4 - 1; 6 - 2) = (3; 4)$. \\ Therefore, the correct answer is $(3; 4)$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use the compound angle formula for cosine to show that $\cos(180^\circ - x) = -\cos x,$ then find $\cos 150^\circ$. .

Final Answer: $-3/2$. $\cos(180^\circ - x) = \cos 180^\circ \cos x + \sin 180^\circ \sin x = -\cos x.$ So $150^= (180^- 30^) = -30^= -3/2.$

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5; -2; 2)$ passing through $P(-3, -4, -3)$.

Final Answer: $-5x - 2y + 2z = 17$. Working through this vectors step:
$d = \mathbf{n} \cdot P = (-5)(-3) + (-2)(-4) + (2)(-3) = 17$

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

1. Check parallelism: Direction vectors are $\mathbf{d}_1 = (2; 1; 3)$ and $\mathbf{d}_2 = (-1; 2; 1)$. Since $\frac{2}{-1} eq \frac{1}{2}$, they are not scalar multiples, so the lines are not parallel. 2. Equate components: $1 + 2\lambda = -\mu \Rightarrow 2\lambda + \mu = -1 \quad (1)$ $2 + \lambda = 5 + 2\mu \Rightarrow \lambda - 2\mu = 3 \quad (2)$ $-1 + 3\lambda = 4 + \mu \Rightarrow 3\lambda - \mu = 5 \quad (3)$ Add (1) and (3): $5\lambda = 4 \Rightarrow \lambda = \frac{4}{5}$. From.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26.$ Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385,$ $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$ Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2.$ Angle θ between vectors.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Isolate the trigonometric terms: $\sec t = \frac{x-1}{2}, \quad \tan t = \frac{y+2}{3}$. Apply the fundamental Pythagorean identity $\sec^2 t - \tan^2 t = 1$: $\left(\frac{x-1}{2}\right)^2 - \left(\frac{y+2}{3}\right)^2 = 1 \Rightarrow \frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$. This is a horizontal hyperbola with:Center: $(h, k) = (1, -2)$Transverse semi-axis: $a = 2$Vertices: $(h \pm a, k) = (1 \pm 2, -2) = (-1, -2)$ and $(3, -2)$

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$. $||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2, \quad ||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2.$ Step 2: Solve for $\cos\theta$: $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · Foundations Evaluate $\iint_R (2x + y)\, dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

Set up iterated integral: $\int_1^2 \int_0^3 (2x + y)\, dx\, dy = \int_1^2 [x^2 + xy]_0^3 dy = \int_1^2 (9 + 3y)\, dy$ Integrate with respect to y: $\left[9y + \frac{3y^2}{2}\right]_1^2 = \left(18 + \frac{12}{2}\right) - \left(9 + \frac{3}{2}\right) = 24 - 10.5 = 13.5 = \frac{27}{2}$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

Compute the dot product: $\mathbf{u} \cdot \mathbf{v} = (2)(6) + (-3)(8) + (4)(3) = 12 - 24 + 12 = 0$ Since the dot product is exactly 0 and neither vector is the zero vector, $\mathbf{u}$ and $\mathbf{v}$ are strictly orthogonal (perpendicular).

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

$\mathbf{d_1} \cdot \mathbf{d_2} = 2 - 1 - 3 = -2$. $|d_1| = 6$, $|\mathbf{d_2}| = \sqrt{11}$. $\cos\theta = \frac{2}{\sqrt{66}} \Rightarrow \theta = \arccos\left(\frac{2}{\sqrt{66}}\right) \approx 75.7°$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.04 Partial Fractions

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13.$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem**

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD $6cmandtheradiusisUSD3$ cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$.
$y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5 ✓

b. 5

c. 15

d. -1

$\mathbf{u} \cdot \mathbf{v} = (2)(4) + (-3)(1) + (5)(-2) = 8 - 3 - 10 = -5$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4 ✓

b. 2

c. -4

d. 5

Orthogonal vectors satisfy $\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 4k + 3(-2) + (-2)(5) = 0 \Rightarrow 4k - 6 - 10 = 0 \Rightarrow 4k = 16 \Rightarrow k = 4$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.

a. $\frac{\pi}{3}$ **(**60°**)** ✓

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

$\mathbf{u} \cdot \mathbf{v} = 1(0) + 1(1) + 0(1) = 1$. $|\mathbf{u}| = \sqrt{2}$, $|\mathbf{v}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$ ✓

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

$\mathbf{u} \times \mathbf{v} = [2(6) - 3(5), -(1(6) - 3(4)), 1(5) - 2(4)] = [12 - 15, -(6 - 12), 5 - 8] = [-3, 6, -3]$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$ ✓

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

By definition, the cross product produces a vector that is orthogonal to both input vectors, directed by the right-hand rule.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. The vectors are parallel (collinear) ✓

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin\theta = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$ or π, meaning the vectors are parallel.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$ ✓

b. $\sqrt{35}$

c. 14

d. $\frac{\sqrt{59}}{2}$

$\mathbf{a} \times \mathbf{b} = [0(-2) - 1(3), -(2(-2) - 1(1)), 2(3) - 0(1)] = [-3, 5, 6]$. Area $= |\mathbf{a} \times \mathbf{b}| = \sqrt{(-3)^2 + 5^2 + 6^2} = \sqrt{9 + 25 + 36} = \sqrt{70}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the triangle with vertices $A(1,0,0), B(0,2,0), C(0,0,3)$.

a. $\frac{7}{2}$ ✓

b. 7

c. $\sqrt{14}$

d. 3

$\vec{AB} = [-1,2,0]$ and $\vec{AC} = [-1,0,3]$. $\vec{AB} \times \vec{AC} = [6,3,2]$. Magnitude $= \sqrt{36+9+4} = \sqrt{49} = 7$. Triangle area $= \frac{1}{2}(7) = 3.5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is $30°$, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15 ✓

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin(30°) = (6)(5)\left(\frac{1}{2}\right) = 15$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. **The volume of the parallelepiped formed by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$** ✓

b. The surface area of the prism formed by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ represents the volume of the parallelepiped with adjacent edges $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1, 0, 0]$, $\mathbf{b} = [0, 2, 0]$, $\mathbf{c} = [1, 1, 3]$.

a. 6 ✓

b. 3

c. 12

d. 2

Volume $= |\det[[1, 0, 0], [0, 2, 0], [1, 1, 3]]| = |1(2 \times 3 - 0)| = 6$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. **The three vectors are coplanar (lie in the same plane)** ✓

b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

A zero scalar triple product implies zero enclosed volume, meaning the three vectors lie in a single common geometric plane (coplanar).

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$ ✓

b. $-\cos x$

c. $\sin x$

d. $-\sin x$

$\sin(x + 90^\circ) = \sin x \cos 90^\circ + \cos x \sin 90^\circ = 0 + \cos x = \cos x$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has direction $\mathbf{d} = (2;\ -1; 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0° ✓

d. 90°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Line 1: $\mathbf{r} = (-3; 2; -3) + s(0; 0; -3)$. Line 2: $\mathbf{r} = (0; -1; 15) + t(-1; 1; -4)$. Find their point of intersection.

a. $(-5, 1, 5)$

b. $(-3, 2, 3)$ ✓

c. $(-4, 0, 4)$

d. $(-1, 1, 4)$

Working through this vectors step: At $s = -2$ (line 1) and $t = 3$ (line 2), both give the point $(-3, 2, 3)$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Line: $\mathbf{r} = (2; -3; 0) + t(-3; 0; 2)$. Plane: $-x - 3y - 3z = 16$. Find the value of t at the point of intersection.

a. -4

b. -5

c. 0

d. **-3** ✓

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 16 \Rightarrow 7 + t(-3) = 16 \Rightarrow t = -3$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the distance from $P(25, 11, 16)$ to the plane $6x + 2y + 3z = -25$.

a. 24

b. **35** ✓

c. 25

d. 43

Working through this vectors step: $\text{Distance} = \frac{|6(25) + 2(11) + 3(16) - (-25)|}{7} = \frac{245}{7} = 35$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Line 1: $\mathbf{r} = (-2;\ -4; 2) + s(-2; 3;\ -1)$. Line 2: $\mathbf{r} = (0;\ -1; 4) + t(-4; 6;\ -2)$. Are the lines parallel?

a. Yes ✓

b. No

c. They are the same line

d. They intersect at one point

Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the distance from the origin to the plane $3x + 4y - 12z = 52$.

a. 5

b. 6

c. **4** ✓

d. 2

Working through this vectors step: Distance $= \frac{|d|}{|\mathbf{n}|} = \frac{52}{13} = 4$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Given $\tan\theta = 3/4$ with θ in Quadrant I, find $\csc\theta$.

a. 5/3 ✓

b. 4/3

c. 5/4

d. 3/5

The 3-4-5 reference triangle gives $\sin\theta = 3/5$, so $\csc\theta = 1/\sin\theta = 5/3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 The Poisson Distribution

Points $A(-5,\ -1,\ -4)$ and $B(-1, 6,\ -3)$ are given. Find $\vec{AB}$.

a. (6; 6; 3)

b. (5; 5; 0)

c. (4; 7; 1) ✓

d. (2; 6; 0)

Working through this vectors step: $\vec{AB} = B - A = (4; 7; 1)$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Further Integration Techniques

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4; -1; 0)$.

a. 90° ✓

b. 45°

c. 60°

d. 30°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.02 Differential Equations

Find the exact value of $\cos 15^\circ$ using $\cos(45^\circ - 30^\circ)$.

a. $6 + 24$ ✓

b. $6 - 24$

c. $2 + 34$

d. $6 + 22$

$15^= 45^3 0^+ 45^3 0^= 2232 + 2212 = 6 + 24.$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

a. 24

b. 12

c. 3

d. 6 ✓

Applying Vectors and Vector-Valued Functions principles, $|a||b|\sin(\pi/6) = 12 \cdot 1/2 = 6$. Therefore, the verified correct option is 6.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

a. $\mathbf{a}$ only

b. $\mathbf{b}$ only

c. neither

d. **both $\mathbf{a}$ and $\mathbf{b}$** ✓

$\mathbf{a} \times \mathbf{b} \perp \mathbf{a}$ and $\perp \mathbf{b}$.

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